(a) necessity for $S = \{0, 1\}$

h_0h_1	$t \in T(h)$
00	yes
01	no
10	no
11	yes

Base 00 retains t when both coordinates are fixed. Removing h_0 admits 10; removing h_1 admits 01.

(b) no unique least slice

h_0h_1	$t \in T(h)$
00	yes
01	yes
10	yes
11	no

At base 00, either $\{h_0\}$ or $\{h_1\}$ suffices. They are incomparable; their intersection $\varnothing$ does not suffice.